

HACKMYSURU 1.0 • PHASE 1 • CIVIC GOVERNANCE &
CLEAN MYSURU

CivicRoute Mysuru

Auto-routing civic complaints to the right office — even while Mysuru's own boundaries are being redrawn.

TEAM

ZENITH

TEAM ID

HM26-7096

SUB-PROBLEM

Routing

THE PROBLEM

Mysuru's civic backlog is real — and growing

550–600t

of garbage generated across Mysuru every single day

~7 lakh t

of old, untreated waste already sitting in backlog

150+ t/day

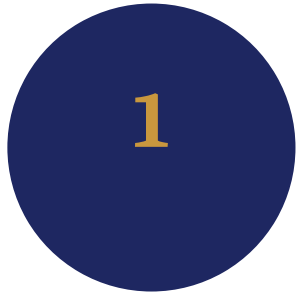
of construction & demolition waste, much of it dumped illegally at night

A pothole, an overflowing bin, or a streetlight complaint can bounce between offices — or vanish — because it's genuinely unclear who owns it right now.

WHY ROUTING FAILS TODAY

The map itself is changing mid-year

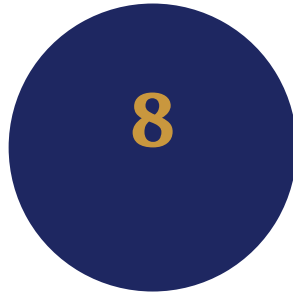
Mysuru City Corporation is absorbing:



City Municipal
Council



Town
Panchayats



Gram
Panchayats

Even local leaders publicly disagree on which villages belong where.

If a routing system assumes a fixed, unchanging map of “who owns what,” **it is already wrong.**

The twist we built for: a routing system that can update itself the moment a jurisdiction shifts — not one that has to be rebuilt.

Three people, one broken handoff



Resident

Situation

Lives near an MCC-panchayat boundary

Needs from us

Report once, in Kannada or English, and see exactly who owns it



Ward / Panchayat Officer

Situation

Inherits new wards mid-year with no updated records

Needs from us

A queue that's already filtered to what's genuinely theirs



Boundary Administrator

Situation

Must reflect authority changes as they're announced

Needs from us

A simple way to update a boundary without a new app release

OUR SOLUTION

One report in, the right office out

A citizen's report is matched against a live jurisdiction map and issue-type rules, returning a routing decision with a confidence score — instead of a citizen guessing which office to call.



A map that updates itself, not one we redeploy

STATIC MAP • REJECTED

A fixed ward-to-office lookup table.

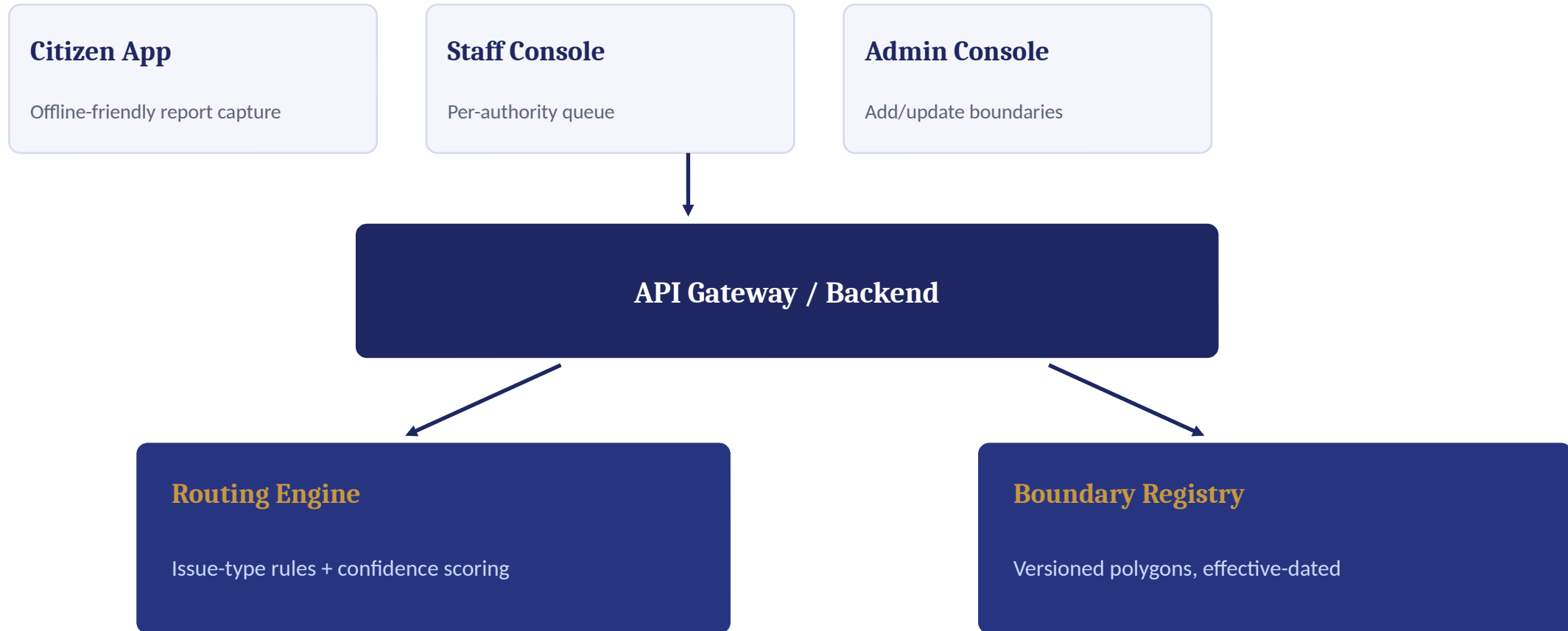
- Fast to build in a 72-hour window
- Breaks the day a single ward changes hands
- Every future boundary change means a code change and a redeploy

VERSIONED BOUNDARY REGISTRY • CHOSEN

Every boundary polygon carries an effective-from date.

- Routing always looks up the boundary version active on the report's date
- A new panchayat absorption = an admin adds a record, not a code change
- Boundary-edge cases get a confidence score and a human check instead of a silent wrong answer

How a report becomes a routing decision



Bad input and no signal are the normal case

Trust, but verify

- Duplicate reports at the same place & time are flagged, not silently accepted
- Photos and locations that don't match the claimed issue are caught before routing
- Anonymous reporting stays open — verification happens on the data, not the identity

Works without signal

- A report made with no network queues locally instead of failing
- Queued reports sync automatically the moment connectivity returns
- Demonstrated live: airplane mode on, report queued, reconnect, sync

What we chose, what we gave up

Dimension	Versioned registry (chosen)	Static lookup table (rejected)
Correct when a boundary shifts	Yes — new record, no redeploy	No — breaks until code changes
Build complexity in 72h	Higher — polygon lookup logic	Lower — a hash-map lookup
Boundary-edge cases	Confidence score + human check	Silent, possibly wrong routing
Cost we accepted	~150ms slower lookup, admin data-entry step	—

What breaks first at full Mysuru scale

At 65+ MCC wards, adjacent panchayats, and thousands of daily reports during a Dasara-scale spike, unindexed point-in-polygon lookups are the first thing to slow down.

The fix: a spatial index on the boundary registry, so lookup time stays flat as ward count and report volume grow.

Thank you

Team ZENITH · HM26-7096 · Sub-problem: Routing

github.com/LEHANYA2023/civicmysuru-hackathon-2026